Question 1

Given:
$\forall x \exists y(C(x) \rightarrow (D(y) \wedge L(y,x)))$
$\forall x \forall y(D(x) \wedge S(y) \rightarrow \neg L(x,y))$
To prove: $\forall x(C(x) \rightarrow \neg S(x))$

Number	**Step**	**Reason**
1	$\forall x \exists y(C(x) \rightarrow (D(y) \wedge L(y,x)))$	Premise(Given)
2	$\forall x \forall y(D(x) \wedge S(y) \rightarrow \neg L(x,y))$	Premise(Given)
3	$\exists y(C(a) \rightarrow (D(y) \wedge L(y,a)))$	Universal Instantiation from 1, a is an arbitrary element
4	$C(a) \rightarrow (D(b) \wedge L(b,a))$	Existential Instantiation from 3, for some b
5	$D(b) \wedge L(b,a)$	Modus Ponens from 4, assuming C(a) is true
6	$L(b,a)$	Simplification from 5
7	$D(b)$	Simplification from 5
8	$D(b) \wedge S(a) \rightarrow \neg L(b,a)$	Universal Instantiation from 2,$x = b$, $y = a$
9	$\neg(D(b) \wedge S(a))$	Modus Tollens from 6, 7 and 8
10	$\neg D(b) \vee \neg S(a)$	De Morgan's law
11	$\neg S(a)$	Disjunctive Syllogism from 7 and 10
12	$C(a) \rightarrow \neg S(a)$	Conditional Proof from 5 and 11
13	$\forall x(C(x) \rightarrow \neg S(x))$	Universal Generalization from 12

Question 2

Solution: Let P(n) be the propostion that $7^n - 6n - 1 \equiv 0 \pmod{36}$, for $n \in \mathbb{Z}, n \geq 1$.
BASIS STEP:
P(1)is true because

$$\begin{aligned} 7^1 - 6(1) - 1 = 7 - 6 - 1 = 0 \\ 0 = 0 \pmod{36} \end{aligned}$$

INDUCTIVE STEP:
For the inductive hypothesis, we assume that P(k) is true for an integer k that $k \in \mathbb{Z}$ and $k \geq 1$. That is, we assume that

$$7^k - 6k - 1 \equiv 0 \pmod{36}$$

for a integer $k \in \mathbb{Z}$ and $k \geq 1$.
To carry out the inductive step using this assumption, we assume that P(k) is true, then we must show that P(k+1) is also true, which is

$$7^{k+1} - 6(k+1) - 1 \equiv 0 \pmod{36}$$

Because

$$\begin{aligned} 7^{k+1} - 6(k+1) - 1 &= 7 \cdot 7^k - 6k - 6 - 1 \\ &= 7 \cdot 7^k - 6k - 7 \end{aligned}$$

So, $7 \cdot 7^k - 6k - 7 = 36n$ for an existing $n \in \mathbb{Z}$.
Using the inductive hypothesis, we need to prove that $7^{k+1} - 6(k+1) - 1 \equiv 0 \pmod{36}$:

$$\begin{aligned} 7^{k+1} - 6(k+1) - 1 &= 7(36n + 6k + 1) - 6k - 7 \\ &= 252n + 42k + 7 - 6k - 7 \\ &= 36(7n + k) \end{aligned}$$

Because we have completed the basis step and the inductive step, by mathematical induction, we know that P(n) is true for all $n \in \mathbb{Z}, n \geq 1$. That is $7^n - 6n - 1 \equiv 0 \pmod{36}$, for $n \in \mathbb{Z}, n \geq 1$.

Question 3

Prove:

For $a \in X \setminus \bigcup \mathcal{F}$, we have $a \in X$ and $a \notin \bigcup \mathcal{F}$, by the definition of set difference.
Since $a \notin \bigcup \mathcal{F}$, a is not in any set $Y \in \mathcal{F}$ by the definition of union.
Thus, for every $Y \in \mathcal{F}$, we have $a \in X \setminus Y$.
This implies $a \in \bigcap\{X \setminus Y \mid Y \in \mathcal{F}\}$, by the definition of set difference.
Since $a \in X \setminus \bigcup \mathcal{F}$, by the definition of intersection,

$$X \setminus \left(\bigcup \mathcal{F}\right) \subseteq \bigcap\{X \setminus Y \mid Y \in \mathcal{F}\}.$$

For $b \in \bigcap\{X \setminus Y \mid Y \in \mathcal{F}\}$, for every $Y \in \mathcal{F}$, we have $b \in X \setminus Y$ by the definition of intersection. Which means, $b \in X$ and $b \notin Y$ for every $Y \in \mathcal{F}$.
Since b is not in any $Y \in \mathcal{F}$, $b \notin \bigcup \mathcal{F}$, by the definition of union.
Since $b \in X$ and $b \notin \bigcup \mathcal{F}$, we have $b \in X \setminus \bigcup \mathcal{F}$.
Because $b \in \bigcap\{X \setminus Y \mid Y \in \mathcal{F}\}$, by the definition of set difference,

$$\bigcap\{X \setminus Y \mid Y \in \mathcal{F}\} \subseteq X \setminus \left(\bigcup \mathcal{F}\right).$$

From both inclusions, we can conclude that:

$$X \setminus \left(\bigcup \mathcal{F}\right) = \bigcap\{X \setminus Y \mid Y \in \mathcal{F}\}$$

Question 4

1. Reflexivity
   Reflexivity is that for every $A \in \mathcal{P}(X)$, $(A, A) \in R$.
   Since the power set $\mathcal{P}(X)$ contains the empty set $\emptyset$ as an element, when $A = \emptyset$, we have $\emptyset \cap \emptyset = \emptyset$. This does not satisfy the condition $A \cap A \neq \emptyset$.
   Therefore, $(\emptyset, \emptyset) \notin R$.
   The relation R is not reflexive.

2. Symmetry
   Symmetry is that for every $A, B \in \mathcal{P}(X)$, there is also $(B, A) \in R$.
   Assume $(A, B) \in R$, which means $A \cap B \neq \emptyset$. Since that $A \cap B = B \cap A$, if $A \cap B \neq \emptyset$, then$B \cap A \neq \emptyset$ is also ture. By the definition of R, the condition $B \cap A \neq \emptyset$ implies that $(B, A) \in R$.
   The relation R is symmetric.

3. Antisymmetry
   Antisymmetry is that for every $A, B \in \mathcal{P}(X)$, whenever $(A, B) \in R$ and $(B, A) \in R$, it follows that $A = B$.
   We can proof that R is not antisymmetry by giving a counterexample of this.
   Let $X = \{0, 1, 2\}$(which is not empty), $A = \{0, 1\}$ and $B = \{1, 2\}$.

   $$A \cap B = \{0, 1\} \cap \{1, 2\} = \{1\}$$

   Since $A \cap B \neq \emptyset$, $(A, B) \in R$.

   $$B \cap A = \{1, 2\} \cap \{0, 1\} = \{1\}$$

   Thus, $B \cap A \neq \emptyset$, $(B, A) \in R$. However, $A \neq B$. So, according to the definition, the relation R is not antisymmetric.

4. Transitivity
   Transitivity is that for every $A, B, C \in \mathcal{P}(X)$, whenever $(A, B) \in R$ and $(B, C) \in R$, it follows that $(A, C) \in R$. We can proof that R is not transitivity by giving a counterexample of this.
   Let $X = \{1, 2, 3, 4\}$, $A = \{1, 2\}$, $B = \{2, 3\}$, and $C = \{3, 4\}$.

   $$A \cap B = \{1, 2\} \cap \{2, 3\} = \{2\} \neq \emptyset$$

   So, $(A, B) \in R$.

   $$B \cap C = \{2, 3\} \cap \{3, 4\} = \{3\} \neq \emptyset$$

   So, $(B, C) \in R$.

   $$A \cap C = \{1, 2\} \cap \{3, 4\} = \emptyset$$

   Since the intersection is the empty set, $(A, C) \notin R$. So, according to the definition, The relation is not transitive.

Question 5

(a)
Let
The probability that an attack event occurs: $P(A) = 0.02$
The probability that no attack event occurs:$P(\overline{A}) = 1 - P(A) = 0.98$
The probability that an event that the IDS raises a high-priority alert occurs:$P(R)$
The probability that an event that the IDS does not raise a high-priority alert occurs:$P(\overline{R})$
Using Bayes' Theorem:

$$P(A \mid \overline{R}) = \frac{P(\overline{R} \mid A) \cdot P(A)}{P(\overline{R})}$$

Since that

$$P(\overline{R} \mid A) = 1 - P(R \mid A) = 0.02$$

$$\begin{aligned} P(\overline{R}) &= P(\overline{R} \mid A)P(A) + P(\overline{R} \mid \overline{A})P(\overline{A}) \\ &= 0.02 \times 0.02 + 0.95 \times 0.98 = 0.9314 \end{aligned}$$

Thus,

$$P(A \mid \overline{R}) = \frac{0.02 \times 0.02}{0.9314} = \frac{0.0004}{0.9314} \approx 0.00042946$$

(b)
Using Bayes' Theorem:

$$P(A \mid R) = \frac{P(R \mid A) \cdot P(A)}{P(R)}$$

$$\begin{aligned} P(R) &= P(R \mid A)P(A) + P(R \mid \overline{A})P(\overline{A}) \\ &= 0.98 \times 0.02 + 0.05 \times 0.98 = 0.0686 \end{aligned}$$

Thus,

$$P(A \mid R) = \frac{0.98 \times 0.02}{0.0686} = \frac{0.0196}{0.0686} \approx 0.28571429$$

(c)
X is the number of days (out of 30) that the IDS raises an alert. Since days are independent, and there is two possible outcome for this (IDS raises a high-priority alert or not), this is a Bernoulli trial.

The expected value of a random variable X on the sample space S is defined as:

$$E(X) = \sum_{s \in S} p(s)X(S)$$

For a binomial random variable X , this simplifies to:

$$E(X) = n \cdot p$$

For X,$n = 30, p = 0.0686$, so

$$E(X) = 30 \times 0.0686 = 2.058$$

The variance of a random variable X using the Theorem

$$Var(X) = E(X^2) - E(X)^2$$

For a binomial random variable:

$$Var(X) = n \cdot p \cdot (1 - p)$$

Substituting the values:

$$Var(X) = 30 \times 0.0686 \times (1 - 0.0686) \approx 1.917$$

(d)

$$E[Y] = E[\text{Damage Cost}] + E[\text{Responding Cost}]$$

$$E[\text{Damage Cost}] = 150000 \times P(A) = 150000 \times 0.02 = 3000$$
$$E[\text{Responding Cost}] = 5000 \times P(R) = 5000 \times 0.0686 = 343$$

Thus,

$$E[Y] = 3000 + 343 = 3343$$